

Functional sequences, series and spaces – 4

Spaces of functions

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Theoretical minimum.

Definition. $C[a, b]$ with $\|f\| = \max |f|$ is a normed space in which convergence is exactly uniform convergence. It is complete and separable, and its closed unit ball is not compact: in an infinite-dimensional normed space the unit ball is never compact (Riesz). The same set of functions with $\rho(f, g) = \int |f - g|$ is a normed space as well, but it is *not* complete.

Theorem 1 (Banach's fixed point). A contraction of a complete metric space into itself has a unique fixed point, and the iterations converge geometrically from any starting point. Applied in $C[a, b]$ to the map $y \mapsto y_0 + \int_{x_0}^x f(t, y(t)) dt$ it gives Picard's existence and uniqueness theorem for $y' = f(x, y)$ with f Lipschitz in y .

Theorem 2 (Baire). A complete metric space is not a countable union of nowhere dense sets. In $C[0, 1]$ this produces objects rather than excludes them: the nowhere differentiable functions form a residual set (Differentiable functions – 8), and so do the nowhere monotone ones.

Theorem 3 (a basis in $C[0, 1]$). There is a system (φ_n) such that every continuous function expands into a uniformly convergent series $\sum a_n \varphi_n$ — the Faber–Schauder system of tent functions, built from the dyadic piecewise-linear interpolations. The monomials $1, x, x^2, \dots$ are *not* such a system, although their span is dense (Continuous functions – 9).

Fact 1 (which norm to take). The uniform norm controls everything pointwise and is the right one for continuity; the L^1 and L^2 norms ignore sets of small measure and are the right ones for integration and orthogonality.

Convergence in L^1 implies neither pointwise convergence nor boundedness, and the completion of $C[0, 1]$ in these norms is no longer a space of continuous functions.

1. Prove that $C[a, b]$ with the uniform norm is complete and separable, and that its closed unit ball is not compact; then prove Riesz's lemma and deduce that the unit ball of an infinite-dimensional normed space is never compact.
2. Prove Banach's fixed point theorem and deduce Picard's theorem: for f continuous and Lipschitz in y the problem $y' = f(x, y)$, $y(x_0) = y_0$ has a unique solution on a small interval. Show by an example that without the Lipschitz condition uniqueness fails.
3. (Kapasëv 1.91) Construct a basis in the space $C[0, 1]$, that is, a system of functions $(\varphi_n)_{n \geq 1}$ such that every continuous $f: [0, 1] \rightarrow \mathbb{R}$ expands into a uniformly convergent series $f(x) = \sum a_n \varphi_n(x)$.
4. (S.T. Yau CSMC 2018, team) Let $C[0, 1]$ carry the metric $\rho(f, g) = \int_0^1 |f - g|$. Show that $(C[0, 1], \rho)$ is not complete, and construct a complete metric vector space $(W, \tilde{\rho})$ together with an isometric embedding of $(C[0, 1], \rho)$ onto a dense subspace of W .
5. (Kapasëv 1.85) Suppose a linear map $T: C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ commutes with differentiation and with multiplication by x . Prove that T is multiplication by a constant: $T(f) = cf$ for some c and all $f \in C^\infty(\mathbb{R})$.